Joe Kanjirathingal

Data Science and Machine Learning
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SUMMARY

B.Tech Artificial Intelligence student graduating in March 2027, seeking data analyst, data science and machine learning internships. Built a tested SQL warehouse over 110 million e-commerce events and fine-tuned and deployed a transformer model for product matching; project experience in Python, SQL, statistics, NLP classification, forecasting and dashboards, with reproducible evaluation.

EDUCATION

B.Tech in Artificial Intelligence | Sep 2023 - Mar 2027 (expected)
Vidya Academy of Science and Technology, Thrissur, Kerala

TECHNICAL SKILLS

Programming and Data: Python, SQL (window functions, CTEs), DuckDB, Parquet, pandas, NumPy, data validation
Analytics: Excel (SUMIFS, XLOOKUP, PivotTables), cohort and funnel analysis, A/B test design, forecasting
Machine Learning: scikit-learn, PyTorch, Hugging Face Transformers, ONNX, NLP, classification, calibration
Visualization and Applications: Chart.js, Three.js, GeoPandas, Leaflet, FastAPI, Django, React, TypeScript
Databases and Tools: PostgreSQL, SQLite, Supabase, Git, GitHub, Vercel

SELECTED PROJECTS

Retail Analytics Lab - E-commerce Data Warehouse and Analysis | Live site
Oct 2026 | Python, SQL, DuckDB, Parquet, NumPy, SciPy, Excel
- Built a reproducible Python and DuckDB pipeline and star schema over 110M e-commerce events (4.6 GB); 14 data tests reconcile rows and $505.1M of revenue from raw files to every output.
- Detected a 48-hour purchase-tracking outage that created a false 53% weekend revenue effect; added automated gap detection and excluded the affected days from conversion analysis.
- Wrote SQL marts (window functions, cohorts, Pareto) and a decision memo: 42% of buyers drive 79% of revenue; 13-day forecast at 5.6% MAPE versus a 5.9% seasonal-naive baseline.

QuoteCheck - Supplier Quotation Review and Product Matching | Project
Oct 2026 - Present | Python, PyTorch, FastAPI, React, PostgreSQL, ONNX, Vercel
- Built a quotation-review app with PDF/XLSX extraction, source-linked corrections, exact pricing and exports.
- Fine-tuned MiniLM on 19,835 product pairs; achieved 74.3% validation average precision and 64.7% F1 on a 4,500-pair WDC Products benchmark, versus 43.4% F1 for the logistic baseline.
- Deployed ONNX CPU inference on Vercel with PostgreSQL; verified by 247 automated tests and 3 browser workflows.

IntentDesk - NLP Classification and Evaluation | GitHub
Oct 2026 | Python, scikit-learn, PyTorch, FastAPI, SQLite
- Built an NLP workflow for 150 intent categories using frozen pretrained MPNet embeddings, a downstream classifier and probability calibration.
- Evaluated 4,500 CLINC150 known-intent test requests: 96.24% accuracy and 0.9624 macro-F1, versus 91.96% accuracy for the lexical baseline; retained human review after unsupported-request evaluation failed.

Kerala Crime Intelligence Map - Data Dashboard | Live demo
Sep 2026 - Present | Python, JavaScript, Leaflet, Chart.js
- Built an interactive dashboard of published Kerala Police statistics with district comparisons, year filters, trend charts, population-adjusted rates and CSV exports.
- Reconciled source totals and distinguished 14-district POCSO coverage from statewide-only categories; documented partial-year 2026 data and population-base limitations.

La Liga - Predictive Modeling and Evaluation | Project
Oct 2026 | Python, PyTorch, NumPy, scikit-learn, SQLite
- Prepared historical football data with source hashes, reconciliation checks and chronological train/test splits.
- Built a leak-free chronological evaluation against a frozen baseline on 86 held-out fixtures; reported that the selected release did not improve on it (47.67% vs 50.00% accuracy).

TRAINING

Python Full-Stack Development | SMEClabs, Kochi | Jun 2026 - Dec 2026 (expected)
Six-month programme covering Python, SQL, Django, HTML/CSS and JavaScript.